SPINE H2-IRL 2023-2024



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Spine H2-IRL

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Project Partners

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1. Introduction

Hydrogen-based technologies have significant potential for the broad decarbonisation of the future Irish energy system, but questions remain regarding security of supply, reliability and system flexibility. There is potential to make more renewable generation more economical leading to even greater deployments of onshore and offshore wind and PV. This will result in new grid challenges and result in greater variability and uncertainty leading to an increased need for flexibility. Potential solutions include long term storage, flexible hydrogen-based generation technologies and batteries. However, the interactions between technologies and their complex interdependencies mean that the problem is not easily modelled.

In this project, Energy Reform extend earlier analysis using an open approach to focus on unanswered questions. Models have been refined and expanded, and new models have been developed, including a new *Reliability Assessment Model*.

The aim of Spine H2-IRL is to develop and publish open models for the comprehensive assessment of a future Irish energy system with widescale deployment of hydrogen production and consumption, along with other potential net zero solutions. This is complimented by analysis using the models to demonstrate their utility and provide useful insights into the future development of the Irish energy system. The detailed models facilitate investment optimisation across different sectors while considering network constraints, long and short-term storage optimisation and a high level of operational detail, with additional models providing more comprehensive flexibility and reliability assessments.

Seven future energy system scenarios, see Section 3, are implemented and evaluated using the models, providing insights and highlighting barriers and opportunities for large-scale hydrogen deployment. Our scenarios focus on a net-zero electricity system with high degrees of electrification, representing a significant step towards a net-zero energy system.

The Spine H2-IRL models, see Section 2, include a *Network Model*, a *Pathways Model*, a *Flexibility Model* and a *Reliability Assessment Model*. High-voltage

transmission (220kV and above) is included in a core *Network Model*, which forms the basis of the other SpineOpt models.

Modelling scope

The aim of Spine H2-IRL is to comprehensively model a future Irish energy system with widescale deployment of hydrogen production and consumption. The detailed models facilitate investment optimisation across different sectors while considering network constraints, long and short-term storage optimisation and a high level of operational detail. Additional models provide more comprehensive flexibility and reliability assessments.

The main focus of the project is to deliver a comprehensive suite of open source modelling tools to address the challenges of the evolving energy system as it moves towards net-zero. A great deal of uncertainty surrounds the emerging hydrogen economy, including costs and demand potential. There is also a lot of uncertainty also surrounding competing sources of flexibility and core input assumptions, including the evolution of electricity demand and realistic capacities of DC interconnection. Ideally, comprehensive sensitivity analysis should be used to address these uncertainties. However detailed sensitivity analysis is beyond the scope of this project, but should be addressed in future work.

As such, the defined scenarios for this work are carefully selected so that the model outputs provide valuable insights for policy makers and industry stakeholders, highlighting opportunities and barriers for large-scale hydrogen utilisation within the future energy system in Ireland. High renewable generation shares and high levels of electrification are considered "no regret" options and part of a robust policy choice for a clean energy future[1]. With this in mind, our scenarios focus on a net-zero electricity system, all with high degrees of electrification and high electricity demands. They represent a significant step towards a net-zero energy system. Although non-electrical

[1] **https://windenergyireland.com/images/files/our-climate-neutral-future-0by50-final-report.pdf**

demands in the building and transport sectors are not explicitly modelled in this work, under the high electricity demand assumptions, a sizable portion of the demand is captured and the decarbonisation facilitated through the net-zero electricity system assumptions.

In the absence of comprehensive sensitivity analysis, and considering the uncertainty surrounding the hydrogen assumptions in particular, Spine H2-IRL have collaborated with the HyLIGHT project through James Carton of DCU. HyLIGHT aims to provide the knowledge, data and the necessary tools to guide the cost-effective decarbonisation and roadmaps for sustainable large-scale implementation of hydrogen technologies in Ireland to enable sector integration for a zero-carbon, secure, resilient energy system[2]. While hydrogen production, via electrolysers, and consumption within the electricity system for power generation is optimised endogenously within the Spine models, HyLIGHT informs hydrogen demand levels outside of the electricity system (primarily industry and heavy transport). Two different hydrogen demand levels are considered, a low demand and high demand, informed by the HyLIGHT scenarios. Likely locations of hydrogen hubs and relative shares of demand are also informed by HyLIGHT. The cost and performance assumptions of hydrogen technologies are aligned, as well as the most viable storage solutions and potential locations. These are included as investment options in the Spine Pathways model.

While great care has been taken to develop these models and tools, time and resources were limited and this should not be considered a full and comprehensive analysis of the future Irish energy system. There are manifold uncertainties particularly surrounding sources of flexibility, the evolution of electricity demand and realistic capacities of DC interconnection. Comprehensive sensitivity analysis should be completed, which is beyond the scope of this project. However, the project demonstrates the usefulness of these open models which are available online.

[2] **https://www.marei.ie/project/hylight/**

Geographical scope and time horizon

All models cover the all-island energy system of Ireland and Northern Ireland. DC interconnectors (existing and planned) are also captured, with price time series for GB and France, generated from a European model based on the ENTSO-E TYNDP dataset[4]. A precise year is not defined for the investment model. The role of hydrogen becomes more relevant at a point where very high shares of renewable generation are possible, and where bulk storage and transport of hydrogen is feasible. As such, our target year is based on a time period towards the end of the 2030s. While most of the models will focus on a single typical year, the reliability model will consider the same target year, with multiple climate years for the renewable generation and demand time series data.

Building on established scenarios

In order to maximise the policy relevance of the modelling results generated in Work Package 4 (Modelling of Green Hydrogen in the Irish Energy Sector), as well as enabling comparability with other studies, Spine H2-IRL draws on established European and Irish scenarios which are consistent with relevant European and national policy directives and assumptions. In particular, Spine H2-IRL scenarios are informed by EirGrid's Tomorrow's Energy Scenarios 2023[3] and ENTSO-E's TYNDP 2024[4], and draw on the recently published National Hydrogen Strategy[5].

EirGrid's Tomorrow's Energy Scenarios 2023 Consultation report was released in December 2023 (The final report, broadly in line with the consultation report was released in 2024, after the Spine H2-IRL scenarios were finalised). The document describes four potential future scenarios for the purpose of

[3] **https://www.eirgrid.ie/site-files/library/EirGrid/Tomorrows-Energy-Scenarios-2023-Consultation-Report.pdf**
[4] **https://2024.entsos-tyndp-scenarios.eu/**
[5] **https://www.gov.ie/en/publication/624ab-national-hydrogen-strategy/**

performing a future System Needs Assessment. The four scenarios are shown in Figure 1. As previously discussed, the Spine H2-IRL scenarios all assume a high level of electricity demand. While the Gas Evolution scenario assumes the development of a hydrogen economy, it also assumes lower demand and slower societal transitions, which does not align with Spine H2-IRL core assumptions. In addition, in the Gas Evolution scenario, uptake of heat pumps and electric vehicles is low with the assumption that hydrogen is widely used for passenger vehicles and heating. However, in Spine H2-IRL we assume that exogenous hydrogen demand is primarily driven by industry and heavy goods vehicles. Spine H2-IRL will primarily draw on the Self-Sustaining scenario, which assumes a high pace of transition.



Figure 1 TES 2023 Scenarios[3]

The European Network of Transmission System Operators for Electricity (ENTSO-E) and the European Network of Transmission System Operators for Gas (ENTSOG) release scenarios for the future European energy system every

two years as part of their respective Ten-Year Network Development Plans (TYNDPs), in accordance with Regulation (EU) 2022/869[6]. The 2024 draft scenarios are currently being finalised and are due to be released by the end of Q1 2024. The draft scenarios are outlined in the TYNDP 2024 Scenario Storylines Report[7] and some high-level data is available on the Visualisation Platform[8]. Three scenarios are developed as part of the TYNDP process: National Trends scenario, which is in line with national energy and climate policies (NECPs) and two deviation scenarios which are described in the storylines report. The Distributed Energy scenario is closely aligned with the TES Self Sustaining scenario in terms of assumed electricity demand.

The National Hydrogen Strategy[5] was published in July 2023. The kick-off date of Spine H2-IRL was chosen to coincide with its publication and ensure alignment on key assumptions. The report provides valuable insights for the creation of credible scenarios. Key end-uses and timelines are outlined in the report along with important infrastructure options and timelines for their implementation. The report informed the selection of the target timescale and shaped the demand assumptions and technology options. More details will be provided on the Spine H2-IRL scenarios and core assumptions in Section 3 of this report.

[6] https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32022R0869&qid=1654587354725&from=en
[7] https://2024.entsos-tyndp-scenarios.eu/wp-content/uploads/2023/12/ENTSOs_TYNDP_2024_Scenarios_Storyline_Report_2023-12-04.pdf
[8] https://2024.entsos-tyndp-scenarios.eu/visualisation-platform/

2. Spine H2-IRL Models

The Spine H2-IRL scenarios outlined in Chapter 3 feed in to Work Package 4, Modelling of Green Hydrogen in the Irish Energy Sector. The open models which are used to generate the results in Work Package 4, and published as part of Work Package 5 (Open Approach, Communication and Dissemination) are developed as part of Work Package 3 (Open Model and Tool Development). The results are described in Chapter 4, and the open models are available online. More detailed descriptions of the models can be found in the Work Package 3 deliverable reports. An overview is provided in this chapter. A workflow for the Spine H2-IRL modelling tasks can be seen in Figure 2.

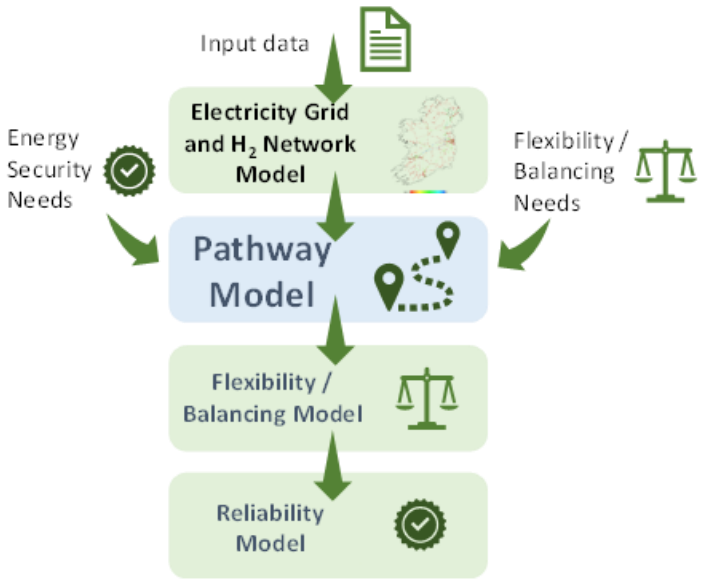


Figure 2 Spine H2-IRL Workflow

The flexible structure of Spine allows us to use a common database for the various models required for this work. The models have been combined and linked in a workflow using Spine Toolbox[9]. Input data includes fuel costs and investment costs and parameters for various generation and network

[9] **https://github.com/spine-tools/Spine-Toolbox**

(electricity and hydrogen) investment options, along with different types of storage. Wind and solar availability and demand (both hydrogen and electricity) times series are also included. For each of the models, the temporal resolution and the level of detail included is selected by the user via alternatives and scenarios within the Spine framework. The inclusion of the transmission network can easily be turned on and off, along with other modelling decisions like the inclusion of transmission investments, integer variables for investment decisions and online status, different technology types and the optimisation of long duration storage. System level constraints can easily be included / excluded and alternative demand levels selected.

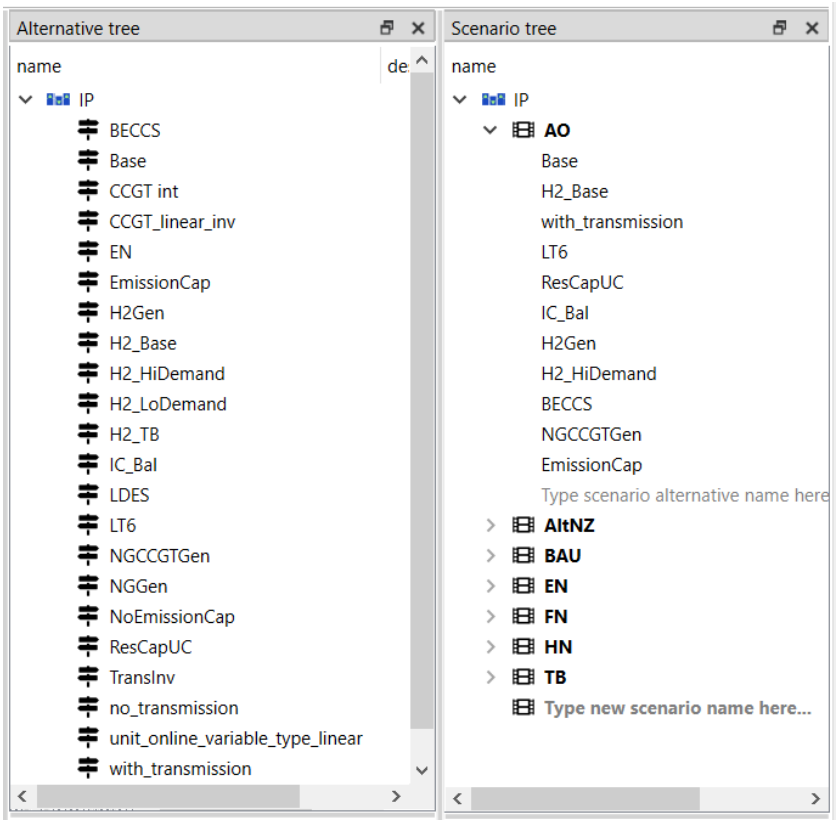
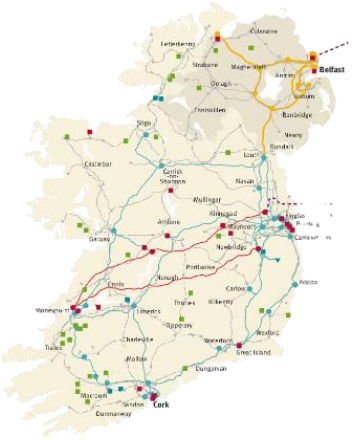


Figure 3 Alternative and Scenario trees, showing a selection of alternatives which can be used to create scenarios.

Network Model

A core *Network Model* has been created, with high voltage electricity transmission that forms the base network for all scenarios. A base generation portfolio common to all scenarios (e.g. hydro / PHS / waste / wind / solar) will is also included. The core Network Model will feed in to the *Pathways Model*, an investment model which will optimise total costs (investments and operating costs) for the different Spine H2-IRL scenarios.



Pathways Model

From the core *Network Model*, a pathway investment model is created. Flexibility requirements (e.g. reserves / inertia floor) are included along with energy security requirements (e.g. a generation capacity reserve). The *Pathways Model* is run for each of the 7 scenarios. For these investment model runs, suitable temporal resolution and decomposition techniques are selected to include as much detail as possible, while managing the model size. Due to the high shares of variable renewable generation it is important to include high levels of both temporal and operational detail in order to adequately capture the flexibility requirements of the system. Thanks to effective collaboration with the Mopo project[10] and TNO, Netherlands, SpineOpt[11] was developed to implement advanced blended representative periods using TulipaClustering[12], allowing greater detail to be captured whilst also capturing long term seasonality within a manageable model size, co-optimising investments, long term storage and detailed operations. Different combinations and quantities of representative days were tested and compared to a simplified model (at a 6 hour resolution) solved for a year all at once. Investment decisions and costs were found to be closely matched. The

[10] **tools-for-energy-system-modelling.org**
[11] **github.com/spine-tools/SpineOpt.jl**
[12] **github.com/TulipaEnergy/TulipaClustering.jl**

trajectory and capacity of the salt cavern were also well captured using the representative periods. 12 representative days were selected using TulipaClustering, whereby the unrepresented days are included as a blend of the 12 representative days.

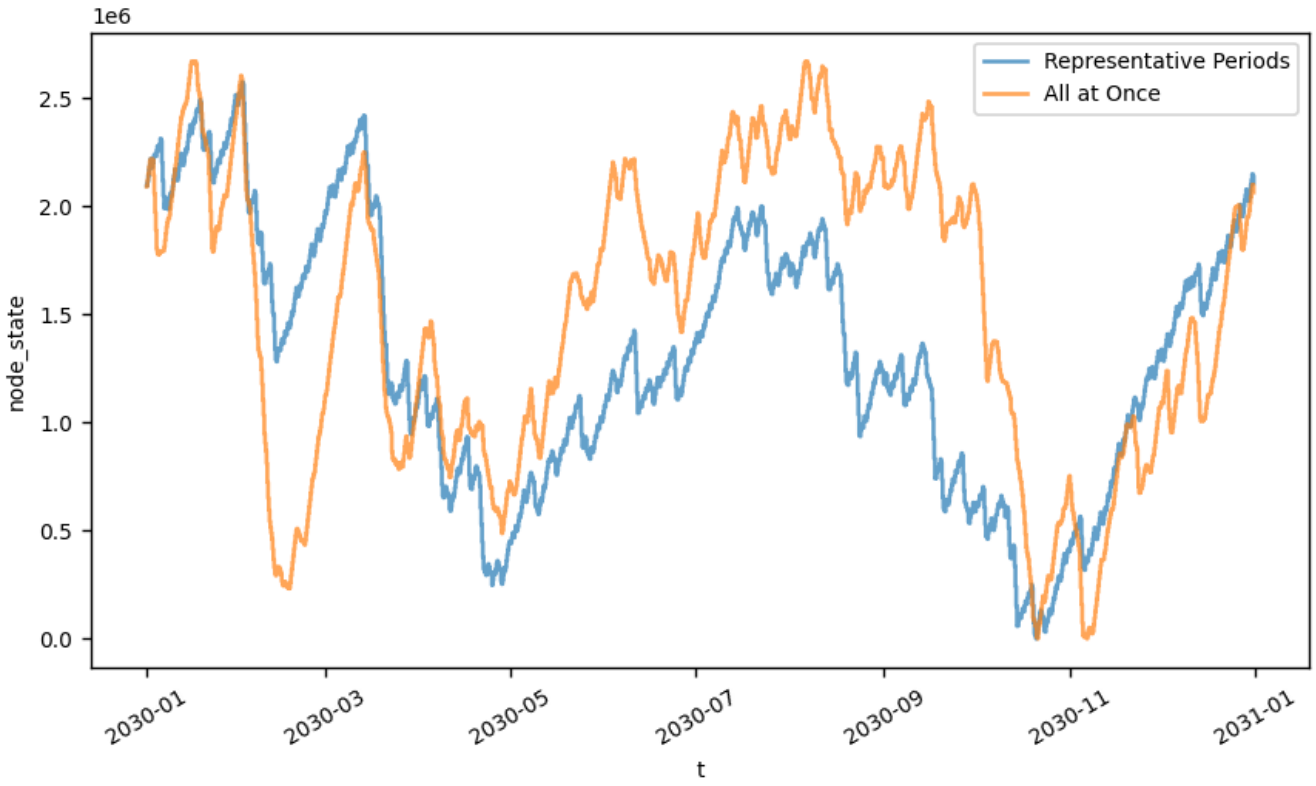


Figure 4 Comparison of the salt cavern hydrogen storage in a simplified model with and without representative periods

Flexibility Assessment Model

The investment decisions from the *Pathways Model* will be fed in to a *Flexibility Model*. While the intention is to include as much temporal and operational detail as possible in the *Pathways Model*, in order to manage the model size some trade-offs are necessary to achieve reasonable run-times. The *Flexibility Model* considers more detailed operations for a full year, facilitating a more comprehensive assessment of the system's capabilities. At this stage, if a lack of flexibility is identified, there is an option adjust the flexibility / balancing requirements defined for the *Pathway Model*.

Reliability Assessment Model

Assessment of the adequacy of a system to meet demand reliably requires analysis of the system under a large number of different scenarios. To account for variability of renewable resources, the performance of the system over a large number of weather years should be assessed. To account for the stochastic nature of technology variability, a large number of forced outage scenarios need to be considered. The future energy system will be increasingly interconnected meaning that reliability of the electricity system increasingly depends on the operation of other sectors, in particular, the gas system and this needs to be considered. Accounting for the contribution of long-term storage and the associated dependency on other sectors poses and additional challenge, requiring long-term modelling. Multi-sector modelling with consideration of long-term issues over a large number of scenarios is computationally challenging.

Existing approaches and tools are often focussed on the electricity system in isolation, and struggle to account for long-term issues.

SpineOpt is uniquely suited to addressing these challenges owing to its flexible nature. This allows us to construct short term and long-term models from the same data. The sectoral flexibility of SpineOpt allows us to easily consider multiple sectors as described in the Pathway model section, where the gas system is explicitly considered. However, running these models over a large number of scenarios required additional developments to allow a sufficient number of scenarios to be run within an acceptable timeframe.

To address these challenges, Energy Reform has developed a number of new functionalities and features within the SpineOpt energy system modelling framework with Spine-H2-IRL project and the Mopo Horizon Europe project.

Stage Functionality

This functionality was developed in SpineOpt by Energy Reform and allows multiple SpineOpt models to be created simultaneously and pass data in

memory. This allows, for example, a long-term model to pass outputs such as long-term storage trajectories to a lower-level operation model. When this is executed in a loop, the overhead of rebuilding the model each time is avoided. The models are built only once, and SpineOpt efficiently identifies the objective and constraint coefficients which have changed and these are directly modified in the optimisation model to avoid the computational overhead of rebuilding the model.

Monte-Carlo Functionality

This new functionality developed within the project allows multi-level Monte-Carlo scenario data to be defined within a model and its impact on system reliability to be efficiently assessed. SpineOpt computes the cross product of the scenarios and in conjunction with the stage functionality described above, SpineOpt constructs a multi-level/multi-model structure to efficiently execute the scenarios. For this case study, the Monte-Carlo scenario levels are:

- **Weather year**: For each weather year scenario, hourly renewable timeseries data (onshore wind, offshore wind and solar), and demand vary based on underlying weather year data. The long-term model is run once for each weather year scenario and outputs of the model include maintenance outage schedules and long-term storage trajectories. These are then inputs to the lower-level scenario, considered in the rolling, multi-sector operational model
- **Forced outage**: For each forced outage scenario, random outages are generated for resources in the system based on forced outage rate and mean time to repair. Outages have been considered on all generating units, electrolysers, interconnectors and batteries.

It is important to note that the framework is completely flexible and one has complete control over:

- The number of monte-carlo scenario levels
- Which data varies for each monte-carlo level and in what way
- Which scenario data is associated with which optimisation model

Once robust portfolios have been selected and tested in the *Flexibility Model* the portfolios will be run through the *Reliability Model*. The *Reliability Model* considers demand time series and renewable availability time series based on multiple weather years. These time series are used in combination with multiple planned and forced outage patterns to generate reliability metrics for the system. Additional steps account for long-term storage. The model has been developed to utilise existing SpineOpt models, simplified and run in a loop until convergence is achieved. SpineOpt was developed within Spine H2-IRL to include monte-carlo capabilities. In combination with SpineOpt's multi-sector flexibility, this allows a mutli-sector reliability assessment accounting for long-term storage over a large number of weather year and outage scenarios. The tool has been made fully available to the public at the end of the Spine H2-IRL project. Again, at this stage of the workflow there is an option to adjust the energy security requirements, if a capacity deficit is identified.

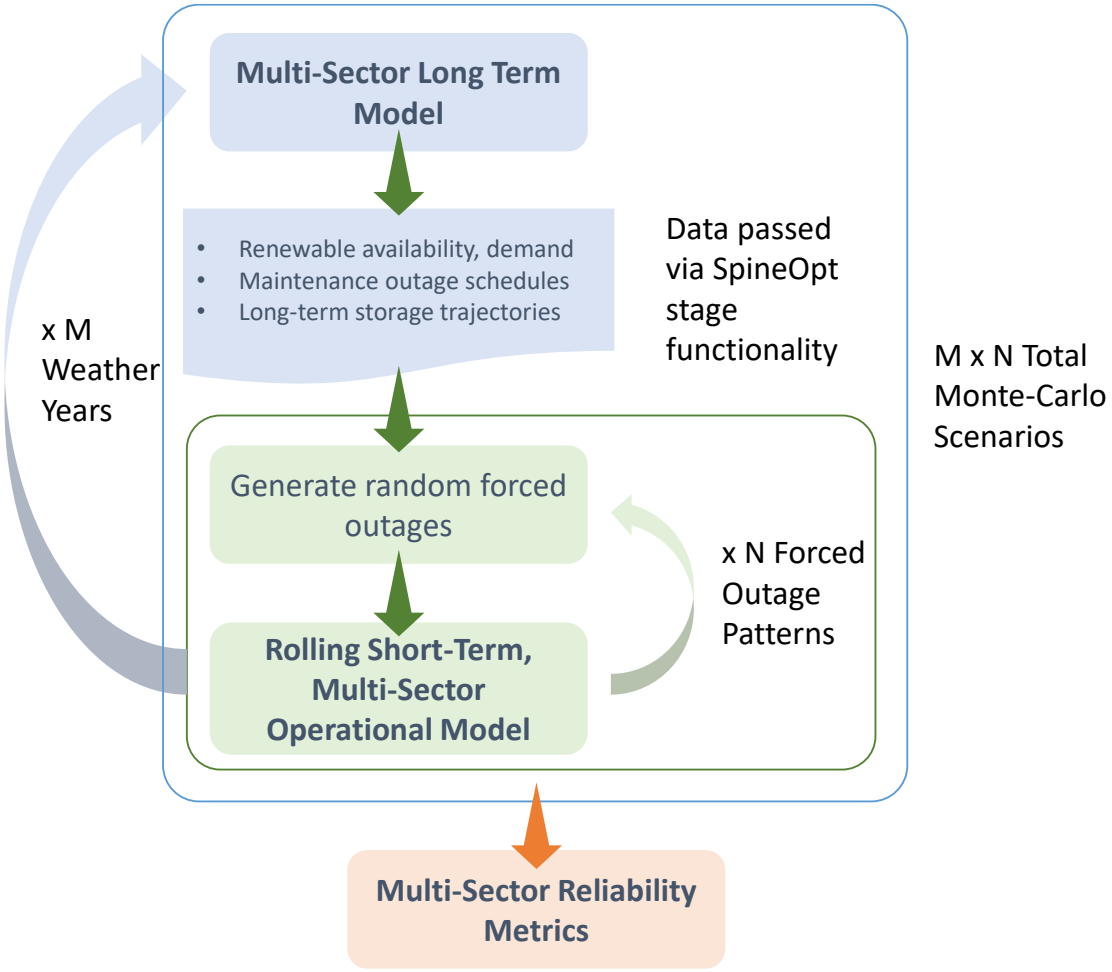


Figure 5: Simplified reliability model structure using nested SpineOpt optimisations

3. Spine H2-IRL Scenarios

As described in Section 1, the Spine H2-IRL scenarios focus on a net-zero electricity system, all with high degrees of electrification and high electricity demands, representing a significant step towards a net-zero energy system. The scenarios are designed to ensure the model outputs provide valuable insights for policy makers and industry stakeholders, highlighting opportunities and barriers for large-scale hydrogen utilisation within the future energy system in Ireland.

Most of the scenarios have a net-zero target for the electricity system. As the base system includes existing waste plant which will generate a low level of emissions, to ensure equal treatment in all scenarios these emissions are handled separately. Carbon costs are applied but the emissions occur outside the net zero constraint. We assume high levels of electricity demand, and the models have the option to invest in hydrogen technologies to assist with overall system balancing. Within the core hydrogen scenarios, different levels of network expansion (electricity transmission network and hydrogen pipelines) are allowed. This highlights the relative importance of the role of network expansion in facilitating optimal deployment of hydrogen technologies and renewable resources. A further hydrogen scenario will take a more optimistic view for the large-scale deployment of hydrogen in the energy system, with lower costs and increased efficiencies for the various hydrogen technologies considered.

Along with the core hydrogen scenarios, three additional scenarios are also included. While two of these scenarios will maintain the net-zero target for the electricity system, a further scenario will consider business as usual, without the net-zero targets, whereby carbon friendly investments is driven by the assumed carbon cost. Details of the 7 Spine H2-IRL scenarios are expanded below.

Scenario Overview

The 7 spine H2-IRL scenarios are summarised in Figure 6 and Table 1. The scenario descriptions is expanded in the remainder of this section.

	Fossil Fuel Generation	Electricity Network Expansion	Hydrogen Network Expansion	Hydrogen Demand	Hydrogen Storage	Hydrogen Investment Costs	Net Zero
Business as Usual	✓	✗	✗	●○○		€€€	✗
Electricity Network	✗	✓	✗	●○○		€€€	✓
Hydrogen Network	✗	✗	✓	●●●		€€€	✓
Full Network	✗	✓	✓	●●●		€€€	✓
Technology Breakthrough	✗	✓	✓	●●●		€€	✓
Alternative Net Zero	✓	✓	✗	●○○		€€€	✓
All Options	✓	✗	✓	●●●		€€€	✓

Figure 6 Spine H2-IRL Scenarios

Business as Usual

The only scenario unconstrained by the net-zero target, the *Business as Usual* scenario still relies on fossil fuel generation, with the assumed carbon price and fossil fuel prices driving investments in alternative supply side solutions. A modest hydrogen demand exists in hubs around Ireland. To meet this demand a level of investment will still be required in electrolysers and hydrogen storage tanks. This scenario is primarily for comparison purposes with the core hydrogen scenarios in terms of costs, and emissions.

Electricity Network

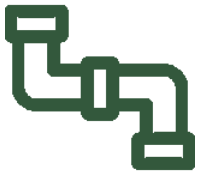
The *Electricity Network* scenario is the first of four core hydrogen scenarios. The net-zero constraint is introduced. However, large-scale infrastructure such as hydrogen pipelines and underground storage are still absent. More expensive and capacity limited tanks can be selected, along with hydrogen fuelled electricity

generation at the designated hydrogen hubs. Electricity network expansion is possible in this scenario.

Table 1Spine H2-IRL Scenario descriptions

Scenario	Description
Business as Usual	**No Net Zero target. Fossil fuel generation remains an option. No further transmission expansion or hydrogen pipelines.**
Electricity Network	**Net Zero target for electricity system. Electricity transmission expansion introduced. Hydrogen investments possible, without large-scale infrastructure.**
Hydrogen Network	**Net Zero target for electricity system. Hydrogen pipeline expansion introduced, along with large-scale underground hydrogen storage.**
Full Network	**Net Zero target for electricity system. Electricity transmission and hydrogen pipeline expansion both facilitated, along with large-scale underground hydrogen storage.**
Technology Breakthrough	**As Full Network, with a further assumption that costs for hydrogen investments have fallen to lower bounds of future estimates.**
Alternative Net Zero	**Net Zero target remains in place, without large scale hydrogen. LDES plays an important role. Limited use of fossil fuels is also permitted, balanced with BECCS.**
All Options	**Net Zero target remains in place, with large scale hydrogen along with limited use of fossil fuels, balanced with BECCS.**

Hydrogen Network



The *Hydrogen Network* scenario has a net-zero constraint enforced. Large-scale hydrogen infrastructure is now included as an investment option. With the possibility of large-scale storage and transport, the higher hydrogen demand is assumed. Hydrogen fuelled electricity generation can be selected both at the designated hydrogen hubs and along the assumed pipeline routes. However, in this scenario electricity transmission expansion is not permitted, which may limit the optimal expansion of both renewable generation and hydrogen production.

Full Network



The *Full Network* scenario has a net-zero target, and expansion of both the electricity transmission system and the hydrogen network is facilitated. In this scenario. the model will be free to co-optimise investments in and locations of renewable generation, hydrogen infrastructure and electricity network capacity expansions. The high level of hydrogen demand is assumed for this scenario.

Technology Breakthrough



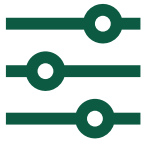
The *Technology Breakthrough* scenario has a net-zero target, and has similar assumptions to the *Full Network* scenario. More optimistic assumptions are used for costs and efficiencies of the key hydrogen investment options. Expansion of both the electricity transmission system and the hydrogen network is facilitated. As with the *Full Network* scenario, the model will be free to co-optimise investments in and locations of renewable generation, hydrogen infrastructure and electricity network capacity expansion.

Alternative Net-Zero



As with the core hydrogen scenarios, the *Alternative Net-Zero* scenario has a net-zero target. However, hydrogen expansion is not considered. The low level of hydrogen demand is assumed for sectors outside the power system, which will require modest investments in electrolysers and hydrogen storage tanks. As hydrogen fuelled electricity generation is not considered, dispatchable generation can be provided by fossil fuel generation in combination with carbon capture and storage (CCS). To balance out the low level of emissions from the CCS plant, negative emission technologies (NETs) will also be considered, namely bioenergy with carbon capture and storage (BECCS), although maximum capacities will be limited. Long duration energy storage (LDES) investment options will also be considered.

All Options



Once again, the net-zero target for the electricity system is applied. As with the core hydrogen scenarios, large scale hydrogen infrastructure is included as investment options and dispatchable generation can be provided by hydrogen fuelled generation. Fossil fuel generation (with or without carbon capture and storage) combined with BECCS to achieve net-zero can also be selected, as with the *Alternative Net Zero* scenario, although LDES is not included as an investment option. The high level of hydrogen demand is assumed.

Base System

A base system is created in the *Network Model*. This consists of network and generation assets common to all scenarios. To create the base electricity transmission network, Future Grid assumptions from EirGrid's ECP constraint forecasts[13] are used. Rather than identify specific network reinforcements, the

[13] **https://www,eirgrid.ie/industry/customer-information/ecp-constraint-forecast-reports**

focus of the network scenarios is to identify the potential order-of-magnitude investments required in network to realize a net-zero system and exploit hydrogen-based technologies and also to explore the potential trade-offs between electricity and gas network investments. Network reduction functionality within the Spine modelling framework is used to aggregate the model to 220 kV+.

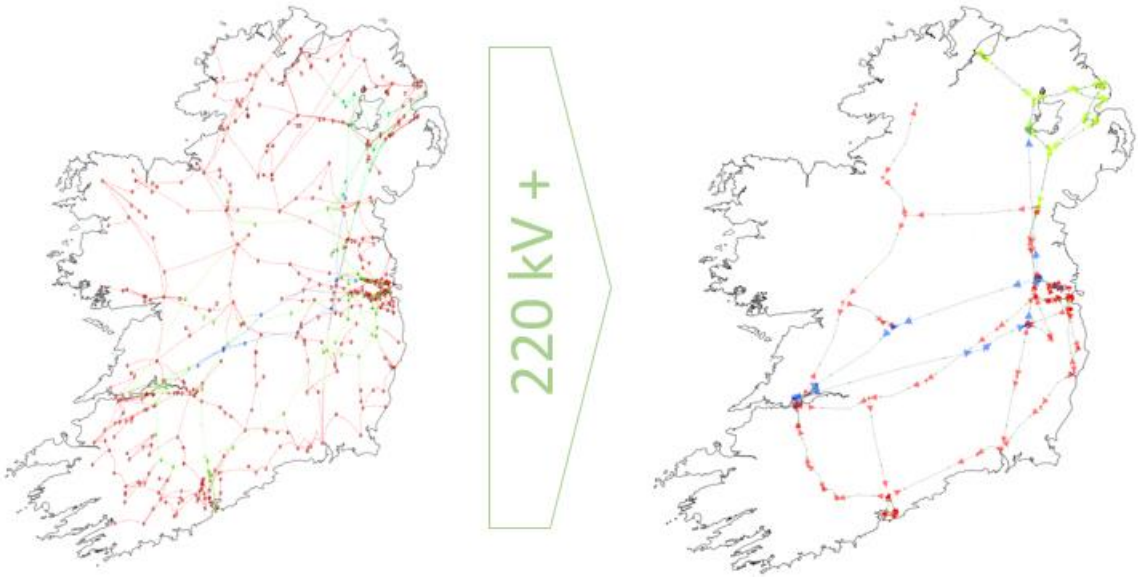


Figure 7 Transmission network pruned to 220 kV+

Existing run-of-river hydro and pumped hydro storage will form part of the base generation portfolio, along with existing waste-to-energy power plants. In addition, a base level of wind and solar generation capacity will be included for all 7 scenarios. The base capacities for wind and solar generation ensure that a realistic distribution of existing / planned generation can be built upon. The base system (prior to investment decisions) includes 9.2, 5.4 & 6.6 GW of installed onshore wind, offshore wind and solar generation capacity.

Generation Technologies

All scenarios have the option to invest in wind and solar generation. The default investment option is for grid connected renewable generation. However, in line with the National Hydrogen Strategy, 2 GW of dedicated offshore wind for

the production of hydrogen is included as an option for the high hydrogen demand scenarios. AC radial connections will be assumed for grid connected offshore wind, although alternative configurations should be considered in future work.

Conventional fossil fuel generation investments are permitted in the *Business as Usual* scenario, the *Alternative Net-Zero* scenario and the *All Options* scenario, which can be coupled with CCS to reduce emissions. Negative emissions technologies (NETs) are included in the *Alternative Net-Zero* scenario and the *All Options* scenario and capacities are capped.

Batteries are available for investment in all 7 scenarios. Batteries of two different durations will be included; 2 hour and 6 hour. Long duration energy storage are an investment option in the *Alternative Net-Zero* scenarios.

Investment Options Summary

A summary of the investment options available in the 7 Spine H2-IRL scenarios can be seen in Figure 8. As some level of hydrogen demand exists in all 7 scenarios, electrolysers and hydrogen storage tanks are included as common investment options. Other investment options common to all 7 scenarios include onshore and offshore wind and solar generation, and grid-scale batteries. Larger scale underground hydrogen storage is included as an option for those scenarios with hydrogen network expansion options, and LDES is considered in the *Alternative Net-Zero* scenario only.
Gas generation can be invested in in the *Business as Usual* scenario, the *Alternative Net-Zero* scenario and the *All Options* scenario, although as the net-zero constraint is enforced in the latter two, investments in CCS may also be required to reduce emissions, and NETs investments will be required in the form of BECCS to counterbalance any emissions from the gas plants.

	Gas CCGT / OCGT	H2 CCGT / OCGT	BECCS	Wind / Solar Gen	Batt (4-8h)	LDES (25-100h)	Elec Net-work	H2 Net-work	Electro lysers	H2 Tank	H2 Under-ground
Business as Usual	✓	✓	✓	✓	✓	✗	✗	✗	✓	✓	✗
Electricity Network	✗	✓	✗	✓	✓	✗	✓	✗	✓	✓	✗
Hydrogen Network	✗	✓	✗	✓	✓	✗	✗	✓	✓	✓	✓
Full Network	✗	✓	✗	✓	✓	✗	✓	✓	✓	✓	✓
Technology Breakthrough	✗	✓	✗	✓	✓	✗	✓	✓	✓	✓	✓
Alternative Net Zero	✓	✗	✓	✓	✓	✓	✓	✗	✓	✓	✗
All Options	✓	✗	✓	✓	✓	✗	✗	✓	✓	✓	✓

Figure 8 Spine H2-IRL Investment Options

For electricity network expansion, it is possible to increase the capacity of the modelled 220kV+ network through investment. This allows the identification of stressed corridors and the order-of-magnitude of reinforcement required. The identification of specific reinforcement options is beyond the scope of this work, although this should be considered in future work.

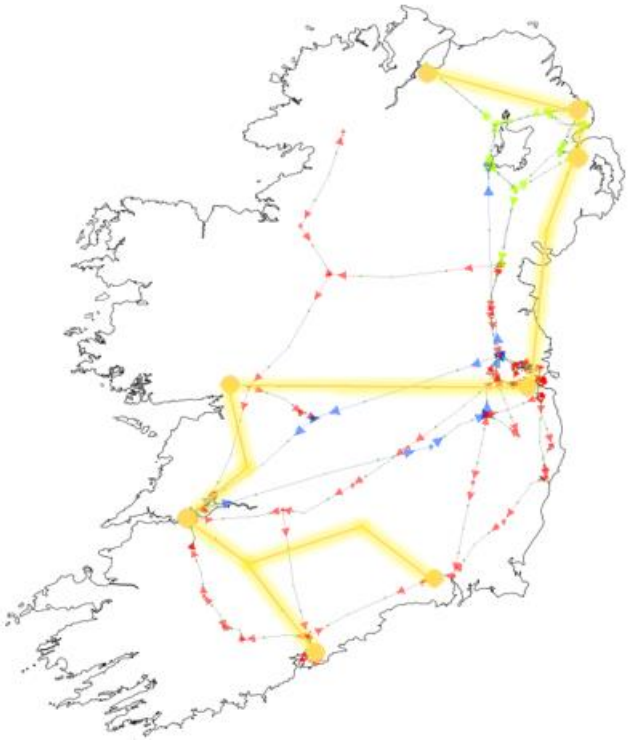


Figure 9 220kV+ electricity network and gas pipeline investment options

The hydrogen network investment options consist of pipelines connecting the assumed hydrogen demand hubs. These hubs are located in Belfast, Larne, Derry, Drogheda, Dublin, Galway, Cork, Waterford and a further hub on the west coast. Generally, the default investment option assumes repurposing of existing gas pipelines. A higher cost option of new pipelines are required for capacities beyond existing capacity available for repurposing. Capacities available for repurposing have been advised by Gas Networks Ireland. Scenarios with hydrogen network investment options also include underground storage in Larne salt caverns as an investment option. Other geological options should be considered in future work.

Hydrogen Assumptions

The key assumptions related to hydrogen production, storage, utilisation and transport for the 7 Spine H2-IRL scenarios are shown in Table 2 below. Sources are listed below the table. Where sources include cost projections based on previous years, an average inflation rate of 3% is assumed. Where projections are not available for the required target year, data is interpolated. As the modelled year is not precisely defined (late 2030s) we use cost and performance projections for 2035 in general, and 2040 for the Technology Breakthrough scenario, assuming technology development and cost reductions are achieved at a faster pace.

Table 2 Key Spine H2-IRL Hydrogen assumptions for the 7 scenarios

		BAU	Elec. Network	H2 Network	Full Network	Tech Breakthr.	Alt. Net-Zero	All Options
Hydrogen Demand	TER	Demand (excluding power gen) informed by HyLIGHT (low demand)		Demand (excluding power gen) informed by HyLIGHT scenario (high demand)			HyLIGHT (low demand)	HyLIGHT (high demand)
Hydrogen Storage Tanks	CAPEX	48.82 €/kWh[c]				18 €/kWh[j]	48.82 €/kWh[c]	
	Efficiency	89%[c]				90%[d]	89%[c]	
Underground Salt caverns	CAPEX			2.37 €/kWh[c]		0.90 €/kWh[j]		2.37 €/kWh[c]
	Efficiency			99%[c/d]				99%[c/d]
Electrolysers	CAPEX	462.5 €/kW[a]				375 €/kW[b]	462.5 €/kW[a]	
	Efficiency	63.75% (LHV)[e]				65.3% (LHV)[f]	63.75% (LHV) [e]	
H2 CCGT	CAPEX		822.5 €/kW[a/b] (+10% assumed for H fuelled turbine)					822.5 €/kW[a/b]
	Efficiency		61%[g]			62%[h]		61%[g]
H2 OCGT	CAPEX		432.5 €/kW[a/b] (+10% assumed for H fuelled turbine)					432.5 €/kW[a/b]
	Efficiency		43%[g]			44%[h]		43%[g]
H2 Pipelines (Repurposed)	CAPEX			EHB Medium price[i]		EHB Low price[i]		EHB Medium[i]
H2 Pipelines (New)	CAPEX			EHB Medium price[i]		EHB Low price[i]		EHB Medium[i]
Offshore Electrolysers	CAPEX			765 €/kW[a]		680 €/kW[a]		765 €/kW[a]
	Efficiency			63.75% (LHV)[e]		65.3% (LHV)[f]		63.75% (LHV)[e]

a/b 20230704-Draft_Supply_Inputs_for_TYNDP_2024_Scenarios.xlsx (2035/2040)
c/d Danish Energy Agency, technology_data_catalogue_for_energy_storage.xlsx (2035/2040)
e/f Danish Energy Agency, data_sheets_for_renewable_fuels.xlsx (2035/2040)
g/h Danish Energy Agency, technology_data_for_el_and_dh.xlsx (2035/2040)
i European Hydrogen Backbone (ehb.eu) – costs depend on pipeline size
j Clean Hydrogen JU - SRIA Key Performance Indicators (KPIs)

Electricity System Assumptions

Table 3 below shows the generation portfolio to be included in the base system. The base system consists of generation capacities common to all scenarios, with renewable generation located at existing and planned project sites, ensuring reasonable final portfolios. Renewable generation expansion will be considered for all scenarios, along with the scenario specific investment options previously outlined. Capacities for the base portfolio have been informed by EirGrid's ECP Future Grid Assumptions and EirGrid's Tomorrow's Energy Scenarios 2023.

Table 3 Base System Generation Portfolio

	Capacity Ireland (MW)	Capacity All-Island (MW)
Pumped Hydro Storage	292	292
Waste	79	94
Run-of-River Hydro	216	216
Onshore Wind	6939	9218
Offshore Wind	4976	5426
Solar	5818	6618
DSU	650	800
DC Interconnectors	2450	3650

Electricity system assumptions, including fuel prices, system constraints and demand levels are shown in Table 4. These assumptions are based on EirGrid's Tomorrow's Energy Scenarios 2023 and ENTSO-E's TYNDP 2024. Note that electrolyser demand will be an output of the model. Other key assumptions related to electricity production and storage for the 7 Spine H2-IRL scenarios are shown in Table 5. Unlike the hydrogen technology assumptions, costs are consistent across the scenarios, although not all technologies are included as investment options for all scenarios. Sources are listed below the table. Where projections are not available for the required target year, data is interpolated.

Table 4 Fuel Prices, Operational Constraints and Demand Levels for Spine H2-IRL Scenarios

	Ireland	Northern Ireland
Carbon Price (€/tonne)	147	
Gas Price (€/GJ)	5.7	
Operational Limit for Inertia (MWs)	20,000 (10,000)**	
Demand TER* (TWh)	79	22
Demand Peak* (GW)	11.3	3.2

*Demand figures exclude electrolysers demand

**10,000 MWs from dispatchable plant

Table 5 Key Spine H2-IRL Electricity technology assumptions for the 7 scenarios

		BAU	Elec. Network	H2 Network	Full Network	Tech Breakthr.	Alt. Net-Zero	All Options
Battery (2 hr)	CAPEX	684 €/kW[a]						
	Efficiency	85%[a]						
Battery (6 hr)	CAPEX	1494 €/kW[a]						
	Efficiency	85%[a]						
LDES (24 hr)	CAPEX						1076 €/kW[b]	
	Efficiency						55%[b]	
LDES (100 hr)	CAPEX						2070 €/kW[b]	
	Efficiency						60%[b]	
CCS	CAPEX	2.15 mill €/tCO2/hr[c]					2.15 mill €/tCO2/hr[c]	
Wind Onshore	CAPEX	1015 €/kW[d]						
Wind Offshore	CAPEX	1725 €/kW[d]						
Solar	CAPEX	355 €/kW[d]						

a NREL Cost Projections for Utility-Scale Battery Storage: 2023 Update, Wesley Cole and Akash Karmakar

b LDES Council Net-zero power Long duration energy storage for a renewable grid

c Danish Energy Agency, technology_data_for_carbon_capture_transport_storage.xlsx

d 20230704-Draft_Supply_Inputs_for_TYNDP_2024_Scenarios.xlsx

4. Results

The results generated by the Spine H2-IRL models are summarised in this Section.

Pathways Assessments

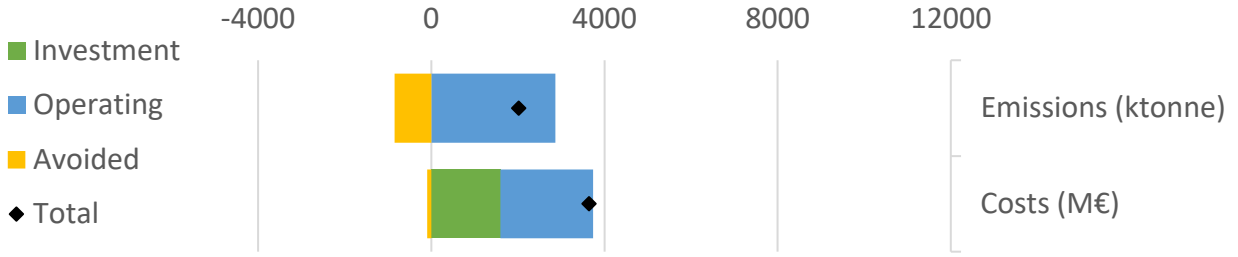
The following section contains the *Pathways Model* results for each of the 7 scenarios. A map is provided showing the locations of the different investments. The electricity transmission network and hydrogen pipelines are also shown, with investments (where considered in the scenario) shown in a heavier line and darker colour.

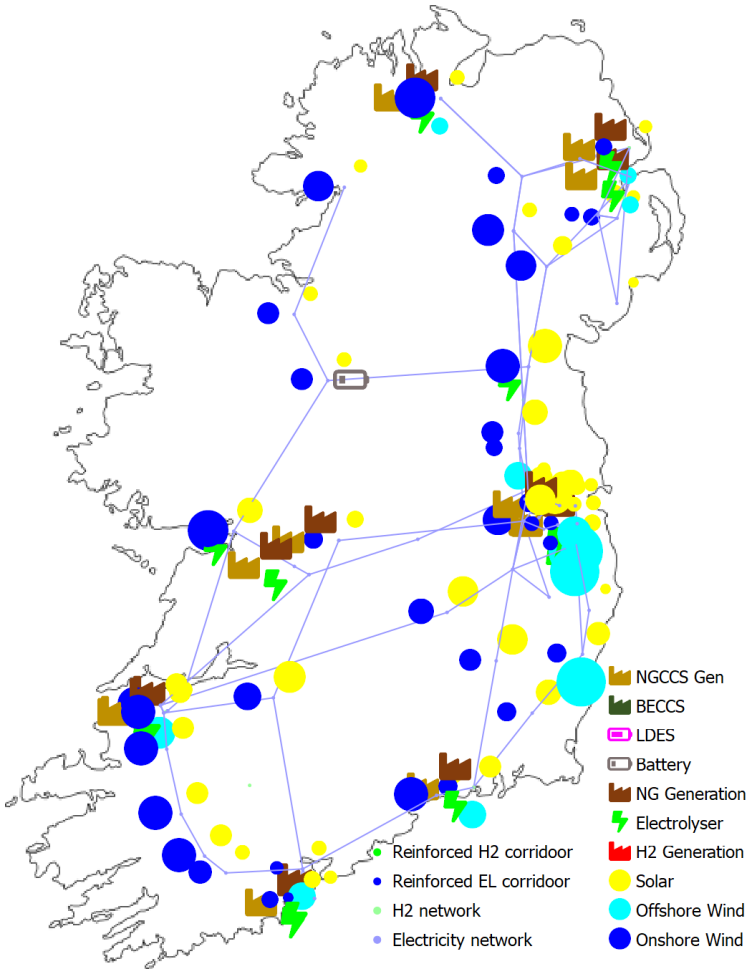
A graph shows the total costs and emissions for each scenario. The emissions and costs are calculated for the power system for the island of Ireland. As the scenarios contain one of 2 different possible levels of hydrogen demand (comprised of demand from industry and transport), avoided emissions and fuel costs in these sectors are included as negative costs / emissions. For the purpose of these calculations, the hydrogen is assumed to primarily displace diesel in heavy transport sector and natural gas in the industrial sector.

A table displays the investment decision for each of the 7 scenarios. For renewable generation, the total capacities (base plus investment) are displayed. A summary and brief discussion of the results for each of the 7 scenarios is also provided.

Business as Usual







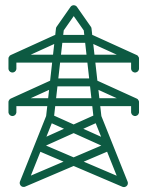
Solar		**10.8 GW**
Onshore Wind		**13.3 GW**
Offshore Wind		**5.4 GW**
H2 Gen		**-**
Electrolysers		**1.0 GW**
H2 Tanks		**3 GWh**
Salt Cavern		
Batteries		**0.13 GWh**
Transmission		
Pipelines		
NG Gen		**5.2 GW**
NGCCS Gen		**5.4 GW**
LDES		
BECCS		**-**

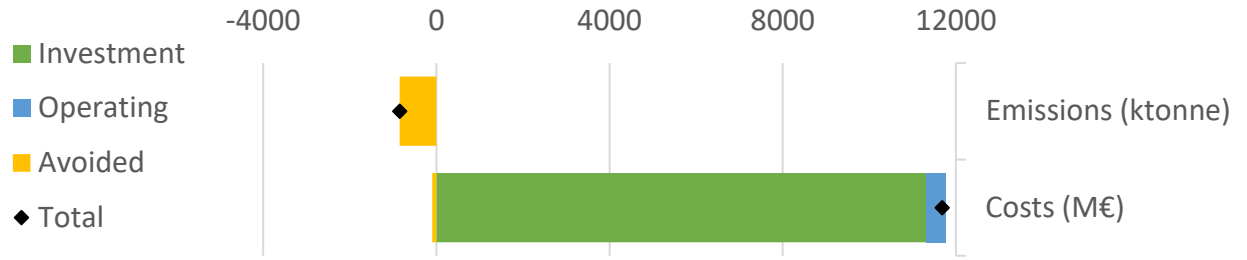
Business as Usual

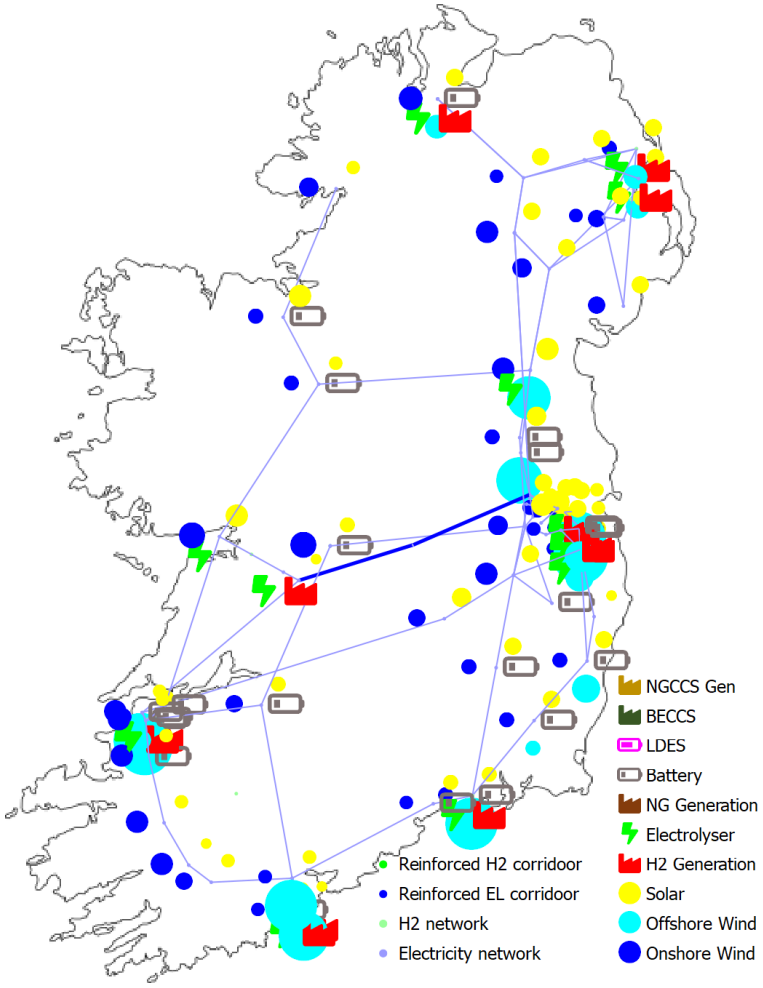


- The carbon price drives investments in low carbon technologies.
- Hydrogen production is sufficent to meet the low Hydrogen demand, but hydrogen fuelled electricity generation is not selected.
- Modest additional investments in renewable generation result in 29.5 GW of installed RES are backed up by almost equal capacities of natural gas fuelled generation and natural gas fuelled generation combined with carbon capture and storage.
- The base portfolio provides most of the flexiblity requirements (flexible demand, interconnectors, PHS & hydro) along with the 1 GW of electrolysers required to meet the low hydrogen demand.
- A small additional capacity of batteries is selected (0.13 GWh) along with 3 GWh of above ground hydrogen storage tanks.
- As low hydrogen demand is included in this scenario, avoided emissions from the industry and transport sectors are modest at 0.85 million tonnes of CO_2.
- Emissions from the electricity generation portfolio amount to 2.86 million tonnes, with a total emissions for the scenario of 2.01 million tonnes of CO_2.
- In the absence of a net zero target, overall costs are low compared to the other scenarios, with annualised investment costs accounting for 43% of the overall costs.
- The hydrogen produced to meet the low demand result in M€100.2 of avoided fuel and carbon costs, giving total annualised costs of M€3,636.
- This scenario demonstrates that a high carbon price justifies significant investments in low carbon technologies, but emissions remain in the absence of a net zero target and there is limited potential for decarbonisation of other sectors.

Electricity Network







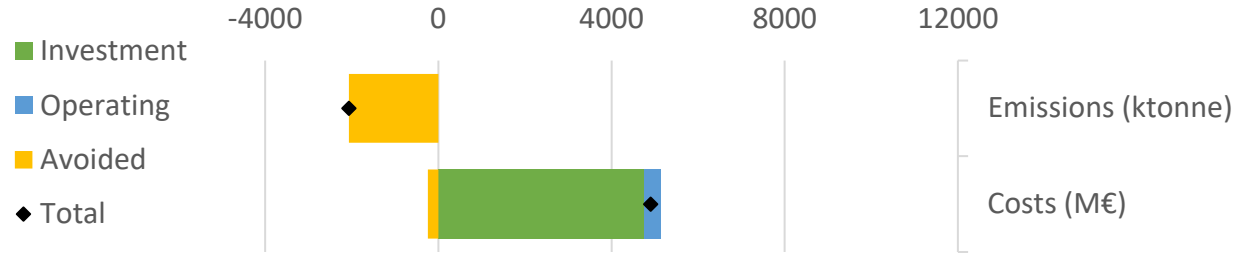
Solar		**15.6 GW**
Onshore Wind		**15.7 GW**
Offshore Wind		**24.1 GW**
H2 Gen		**9.1 GW**
Electrolysers		**14.0 GW**
H2 Tanks		**557.8 GWh**
Salt Cavern		
Batteries		**26.8 GWh**
Transmission		**160.0 km**
Pipelines		
NG Gen		
NGCCS Gen		
LDES		
BECCS		

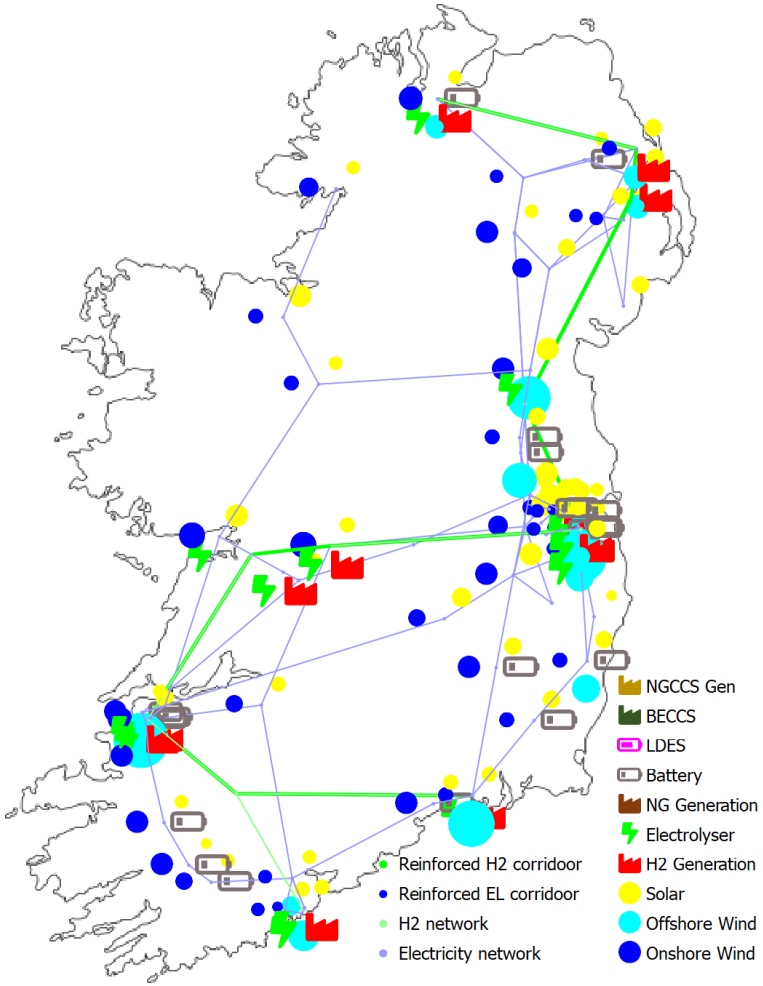
Electricity Network



- While electricity transmission investments can occur, no large scale infrastructure is included in this scenario. A net zero constraint is enforced for the power system, driving investment decisions.
- Hydrogen production is sufficent to meet the low Hydrogen demand, and hydrogen fuelled electricity generation provides most of the dispatchable capacity.
- Large capacities of additional renewable generation result in 55.4 GW of installed RES.
- The large capacity of renewable generation is balanced by investments in grid scale batteries (26.8 GWh).
- Some transmission investments occur, but the model preferentially selects sites for renewable generation and invests in batteries as a lower cost option to wide-scale transmission investments.
- In the absence of large scale hydrogen infrastructure, 558 GWh of above ground hydrogen storage tanks are required.
- As low hydrogen demand is included in this scenario, avoided emissions from the industry and transport sectors are modest at 0.85 million tonnes of CO_2, which results in a small negative total for emissions in this scenario.
- Overall costs are extremely high for this scenario, with annualised investment costs accounting for 96% of the overall costs.
- The hydrogen produced to meet the low demand result in M€100.2 of avoided fuel and carbon costs, giving total annualised costs of M€11,679 (over 3 x Business as Usual).
- This scenario demonstrates that it may be technically possible to achieve a hydrogen based net zero system in the absence of large scale hydrogen infrastructure and low cost storage, but it is a very expensive and inefficient solution.

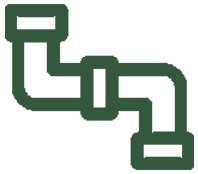
Hydrogen Network





Solar		**15.6 GW**
Onshore Wind		**15.7 GW**
Offshore Wind		**18.4 GW**
H2 Gen		**8.9 GW**
Electrolysers		**11.5 GW**
H2 Tanks		**-**
Salt Cavern		**3.0 TWh**
Batteries		**3.1 GWh**
Transmission		
Pipelines	**Repurposed**	**575km**
	New	**312km**
NG Gen		
NGCCS Gen		
LDES		
BECCS		

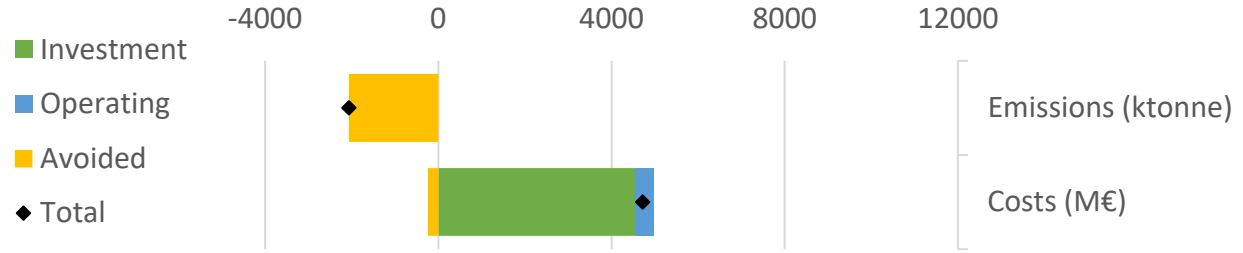
Hydrogen Network

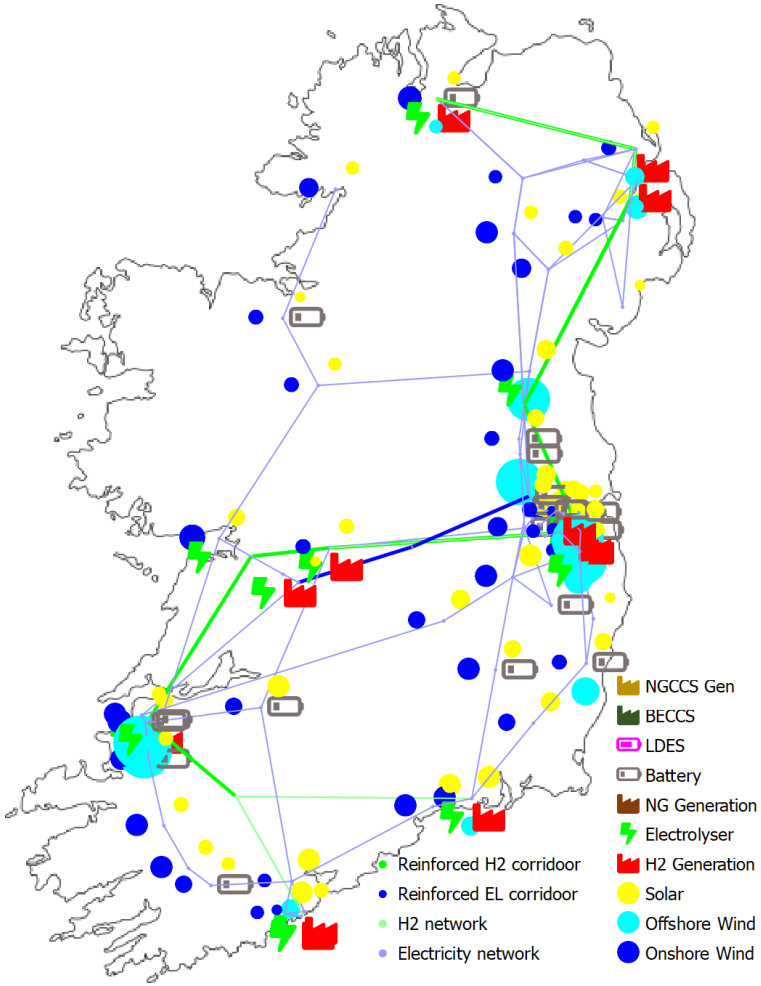


- Large scale gas infrastructure expansion is included in this scenario. A net zero constraint is enforced for the power system, which drives investment decisions.
- Hydrogen production is sufficent to meet the high hydrogen demand, and hydrogen fuelled generation provides most of the dispatchable capacity.
- Significant capacities of additional renewable generation result in 49.7 GW of installed RES, roughly 10% less than in the Electricity Network scenario.
- The expansion of the hydrogen network allows the electolysers capacities to be be more efficiently deployed and dispatched.
- With the introduction of low cost bulk storage (salt cavern), additional hydrogen fuelled generation is favoured over grid scale batteries, with more moderate levels of investments occurring (3.1 GWh).
- The hydrogen network and salt cavern storage displaces more expensive above ground storage.
- As high hydrogen demand is included in this scenario, avoided emissions from the industry and transport sectors are larger at 2.07 million tonnes of CO_2, which results in a more significant negative total for emissions in this scenario.
- Overall costs are significantly reduced compared to the Electricity Network scenario (42% of EN total), with annualised investment costs accounting for 92% of the overall costs.
- The hydrogen produced to meet the high demand result in M€241.8 of avoided fuel and carbon costs, giving total annualised costs of M€4,904 (35% higher than Business as Usual).
- This scenario demonstrates that large cost reductions and efficiencies can be achieved in a net zero hydrogen based system when large scale infrastructure and low cost bulk storage is available.

Full Network







Solar		**15.6 GW**
Onshore Wind		**15.7 GW**
Offshore Wind		**17.4 GW**
H2 Gen		**8.6 GW**
Electrolysers		**11.6 GW**
H2 Tanks		**-**
Salt Cavern		**3.0 TWh**
Batteries		**1.9 GWh**
Transmission		**123.7 km**
Pipelines	**Repurposed**	**458 km**
	New	**253 km**

NG Gen

NGCCS Gen

LDES

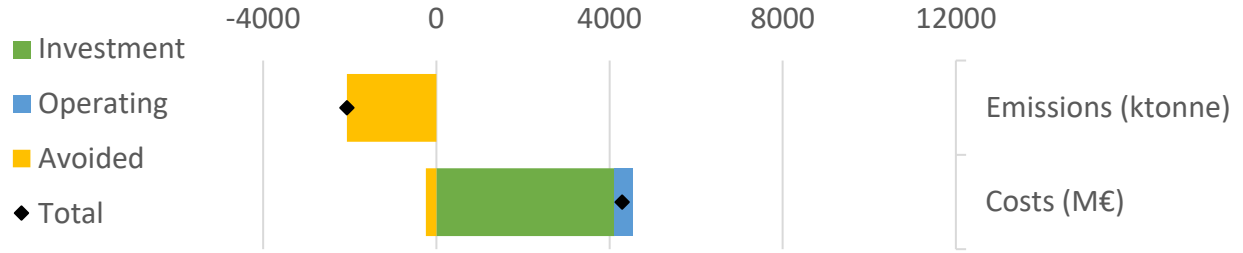
BECCS

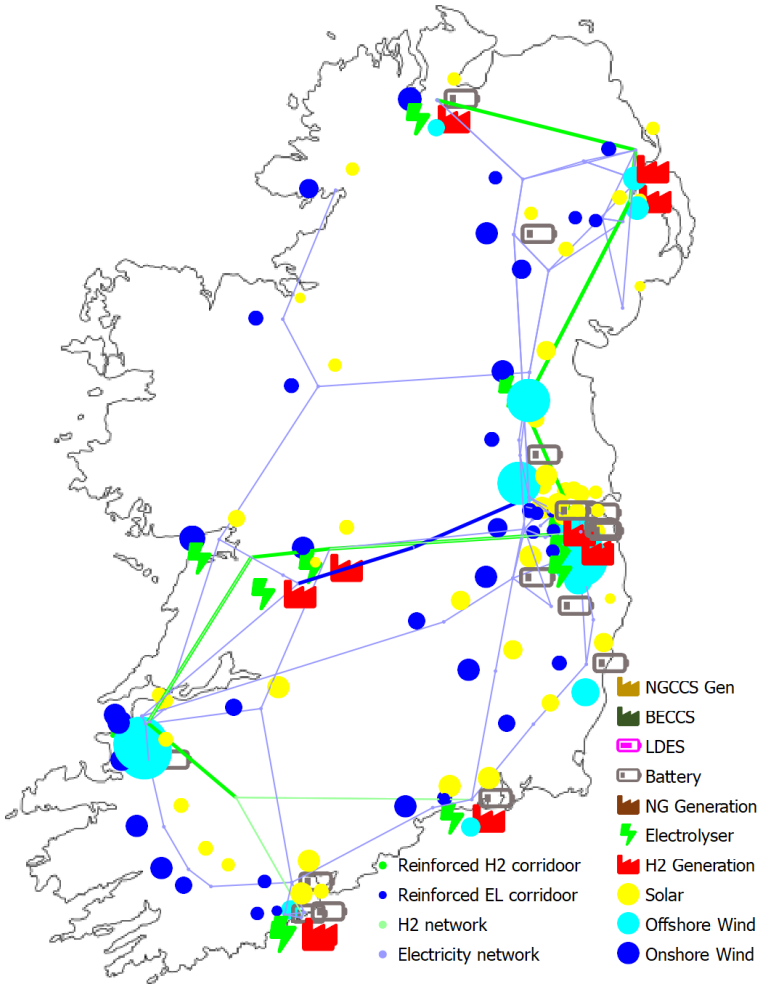
Full Network



- Both electricity transmission and large scale hydrogen infrastructure investments are included in this scenario. A net zero constraint is enforced for the power system, driving investment decisions.
- Hydrogen production is sufficent to meet the high Hydrogen demand, and hydrogen fuelled electricity generation provides most of the dispatchable capacity.
- Significant capacities of additional renewable generation result in 48.7 GW of installed RES, 1 GW less than in the Hydrogen Network scenario.
- A modest investment in electricity transmission capacity occurs, coupled with a small drop in battery investments compared to the Hydrogen Network scenario (1.9 vs 3.1 GWh).
- The hydrogen network and salt cavern storage displaces the more expensive above ground storage tanks, with salt cavern capacity matched to the Hydrogen Network scenario.
- As high hydrogen demand is included in this scenario, avoided emissions from the industry and transport sectors are 2.07 million tonnes of CO_2, which results in a negative total for emissions in this scenario.
- Overall costs are reduced slightly (~ 3%) compared to the Hydrogen Network scenario, with annualised investment costs accounting for 91.5 % of the overall costs.
- The hydrogen produced to meet the high demand result in M€241.8 of avoided fuel and carbon costs, giving total annualised costs of M€4,717 (30% higher than Business as Usual).
- This scenario demonstrates that further efficiencies and cost reductions can be achieved when tranmsission expansion on key lines is facilitated.

Technology Breakthrough





Solar		**15.6 GW**
Onshore Wind		**15.7 GW**
Offshore Wind		**17.6 GW**
H2 Gen		**8.6 GW**
Electrolysers		**12.1 GW**
H2 Tanks		**-**
Salt Cavern		**4.1 TWh**
Batteries		**0.6 GWh**
Transmission		**153.9 km**
Pipelines	**Repurposed**	**458 km**
	New	**292 km**

NG Gen

NGCCS Gen

LDES

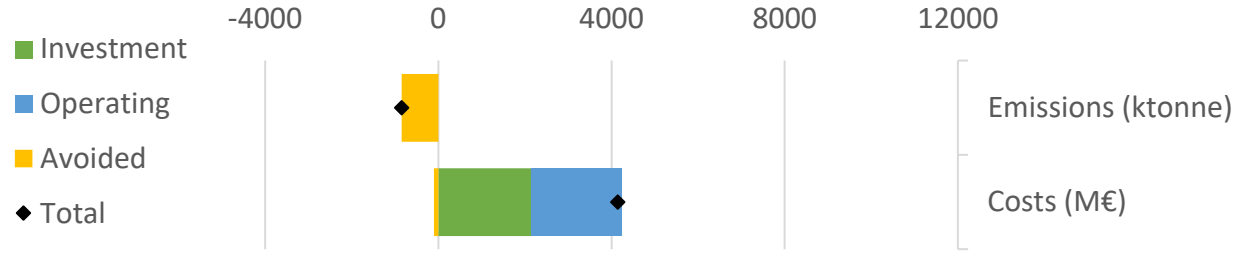
BECCS

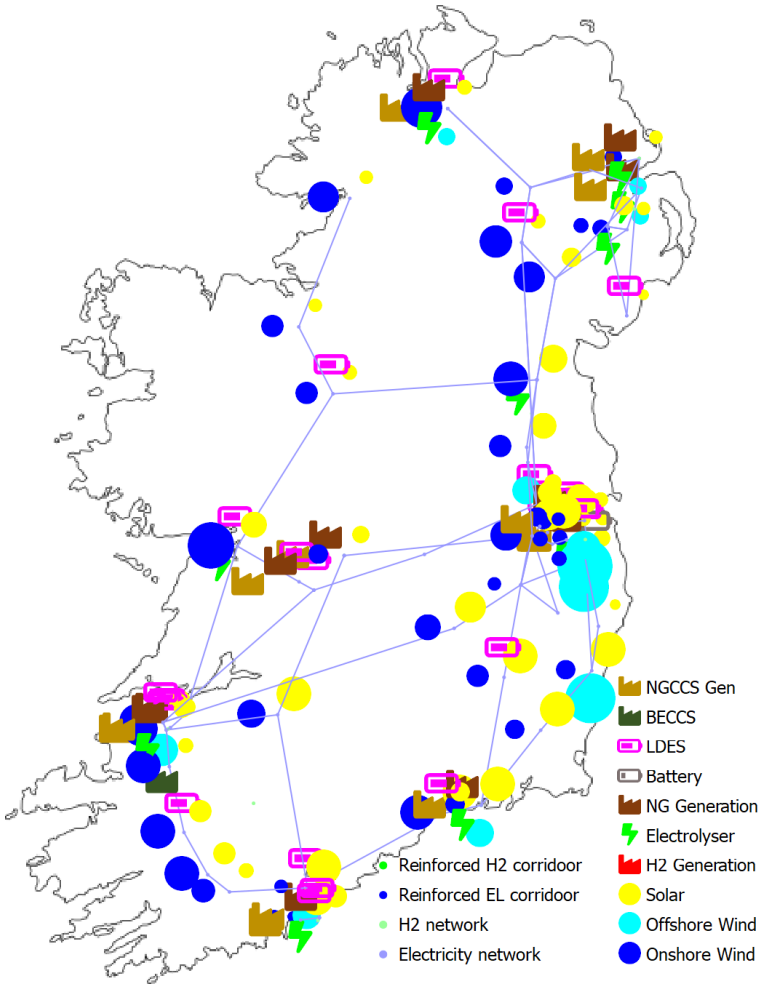
Technology Breakthrough



- Both electricity transmission and large scale hydrogen infrastructure investments are included in this scenario. A net zero constraint is enforced for the power system, driving investment decisions.
- Lower costs and increased efficiencies are assumed for the hydrogen technologies.
- Hydrogen production is sufficent to meet the high Hydrogen demand, and hydrogen fuelled electricity generation provides most of the dispatchable capacity.
- Significant capacities of additional renewable generation result in 48.9 GW of installed RES, similar to the Full Network scenario.
- A modest investment in transmission capacity occurs, coupled with a further drop in battery investments compared to the Full Network scenario (0.6 vs 1.9 GWh).
- Additional investments in both electrolysers (+ 5%) and salt cavern storage (+ 39%) occur compared to the Full Network scenario.
- As high hydrogen demand is included in this scenario, avoided emissions from the industry and transport sectors are 2.07 million tonnes of CO_2, which results in a negative total for emissions in this scenario.
- Overall costs are reduced considerably (~ 9%) compared to the Full Network scenario, with annualised investment costs accounting for 90 % of the overall costs.
- The hydrogen produced to meet the high demand result in M€241.8 of avoided fuel and carbon costs, giving total annualised costs of M€4,291 (18% higher than Business as Usual).
- This scenario demonstrates that if the more ambitious targets are reached for hydrogen technology costs and efficiencies, overall costs move closer to the business as usual scenario, with overall reduced emissions.

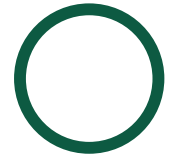
Alterative Net Zero





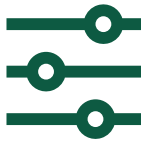
Solar	**13.4 GW**
Onshore Wind	**14.0 GW**
Offshore Wind	**5.4 GW**
H2 Gen	**-**
Electrolysers	**1.0 GW**
H2 Tanks	**3.8 GWh**
Salt Cavern	**0.65 TWh**
Batteries	**0.12 GWh**
Transmission	
Pipelines	
NG Gen	**2.3 GW**
NGCCS Gen	**6.4 GW**
LDES	**47.6 GWh**
BECCS	**300 MW**

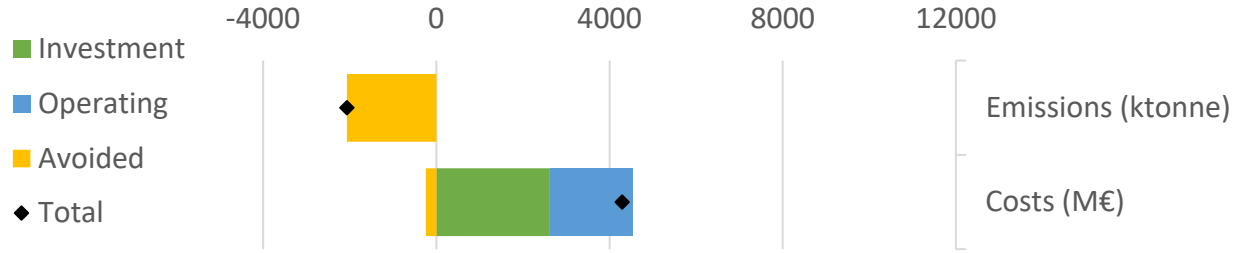
Alternative Net Zero

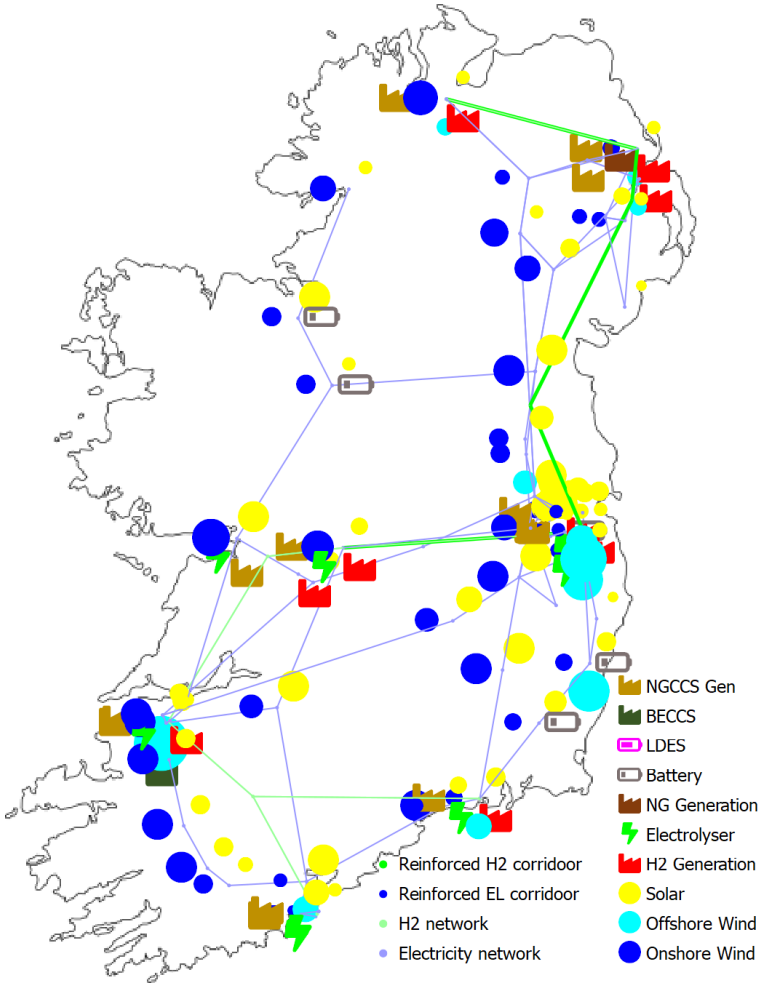


- A net zero constraint is enforced for the power system. While electricity transmission investments are allowed, no large scale hydrogen infrastructure is included.
- Fossil fuel generation (with or without CCS), LDES and. BECCS can all be selected.
- Hydrogen production is sufficent to meet the low Hydrogen demand, while no hydrogen fuelled electricity generation is selected in the absence of large scale infrastructure.
- Modest capacities of additional renewable generation result in 31.7 GW of installed RES, 2.2 GW more than the Business as Usual scenario.
- As with the Business as Usual scenario, the base portfolio and the 1 GW of electrolysers required to meet the low hydrogen demand provides most of the short term flexiblity requirements for the system, with a small capacity of battery investments (0.12 GWh).
- As low hydrogen demand is included in this scenario, avoided emissions from the industry and transport sectors are modest at 0.85 million tonnes of CO_2, which results in a small negative total for emissions in this scenario.
- Overall costs are the lowest of the net zero scenarios, with annualised investment costs accounting for 51 % of the overall costs.
- The hydrogen produced to meet the low demand result in M€100.2 of avoided fuel and carbon costs, giving total annualised costs of M€4,144 (14% higher than Business as Usual).
- This scenario demonstrates that if alternative low carbon technologies become viable, such as carbon capture and storage and alterntive LDES technologies, a lower cost net zero electricity system can be achieved. However, there is uncertaintly surrounding both the LDES and CCS assumptions. In addition in the absence of hydrogen infrastructure there is less potential for the decarbonisation of the wider energy system.

All Options

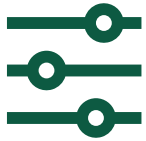






Solar		**15.6 GW**
Onshore Wind		**15.7 GW**
Offshore Wind		**6.6 GW**
H2 Gen		**2.9 GW**
Electrolysers		**5.0 GW**
H2 Tanks		**-**
Salt Cavern		**0.65 TWh**
Batteries		**0.95 GWh**
Transmission		
Pipelines	Repurposed	**344 km**
	New	**13 km**
NG Gen		**0.4 GW**
NGCCS Gen		**5.6 GW**
LDES		
BECCS		**300 MW**

All Options



- A net zero constraint is enforced for the power system with both large scale hydrogen infrastructure and CCS investments allowed.
- Hydrogen production is sufficent to meet the high Hydrogen demand, with a combination of hydrogen fuelled natural gas (primarily with CCS) electricity generation selected.
- Large capacities of additional renewable generation result in 37.9 GW of installed RES, between the Alternative Net Zero and Hydrogen Network scenarios.
- Electrolyser capacities also lie between the Alternative Net Zero and the Hydrogen Network scenarios at 5 GW, meeting the high hydrogen demand some of the hydrogen generation capacity.
- The short term flexiblity requirements for the system increase compared to Alternative Net Zero and an increase in battery capacity is seen (0.95 vs 0.12 GWh).
- As high hydrogen demand is included in this scenario, avoided emissions from the industry and transport sectors are 2.07 million tonnes of CO_2, which results in a negative total for emissions in this scenario.
- Overall costs are comparable to the Technology Breakthrough scenario, with annualised investment costs accounting for 58 % of the overall costs.
- The hydrogen produced to meet the high demand result in M€241.8 of avoided fuel and carbon costs, giving total annualised costs of M€4,291 (18% higher than Business as Usual).
- This scenario demonstrates that a combination of technologies , including large scale hydrogen production, can achive efficient solutions for the power system, while also achieving significant emission reductions in the wider energy system.

Figure 10 below combines the emissions results for the 7 scenarios. For 6 of the 7 scenarios, the net-zero constraint for the power system results in negative emissions once avoided emissions by the displacement of fossil fuels by the produced hydrogen is included. The level of the avoided emissions is driven by the assumed hydrogen demand (high demand or low demand) for the scenario.

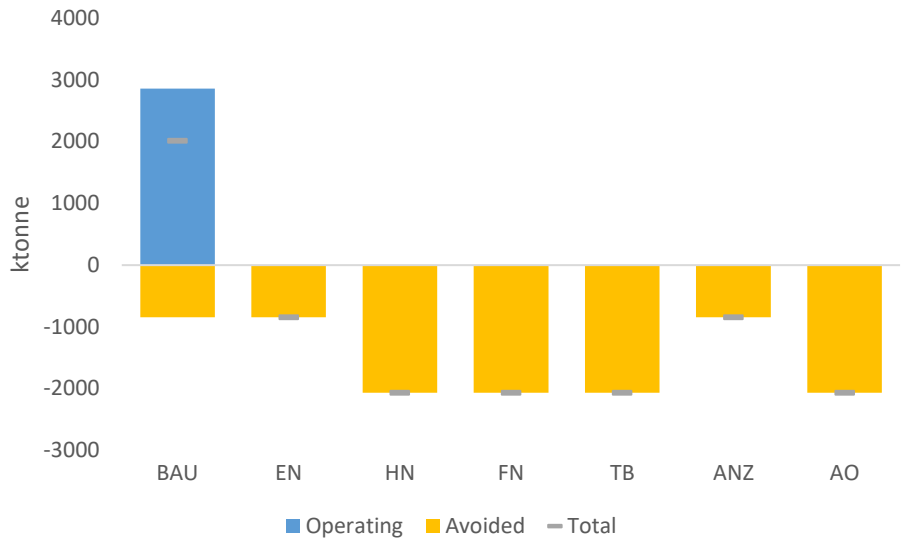


Figure 10 Emissions for the 7 Spine H2-IRL Scenarios

For comparison the annualised costs for the 7 scenarios are shown in Figure 11, with avoided costs from displaced fossil fuels shown as a negative value.

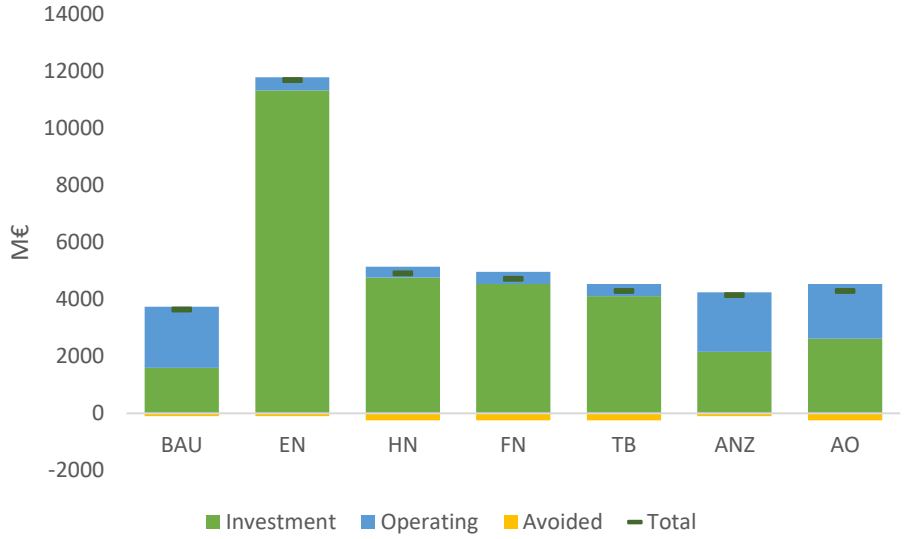


Figure 11 Annualised costs for the 7 Spine H2-IRL scenarios

The Business as Usual scenario results in the lowest overall annualised costs. However, as shown in Figure 10, this is not a net zero solution. Achieving a net zero power system. The Electricity Network scenario can be seen as an outlier. This scenario assumes that hydrogen form the backbone of the electricity system, without any large scale infrastructure or alternative sources of long duration dispatchable capacity. The remaining core hydrogen scenarios all demonstrate significant reductions in overall costs once the hydrogen pipelines and large-scale storage are introduced. The Alternative Net Zero scenario demonstrates the lowest overall costs of the net zero scenarios. However, this scenario assumed the lower demand level for hydrogen beyond the power system and avoided emissions are at a lower level. This scenario also depends on an established carbon capture and storage network. The assumed costs and efficiencies for LDES are also uncertain. The All Options scenario is the next lowest cost option. This scenario also assumes the high demand option for hydrogen and has greater potential for the decarbonisation of the wider system. It also relies on an established carbon capture and storage network. Questions also remain surrounding the reliability of the different portfolios. The reliability model created as part of this project can be used to complete more comprehensive analysis.

Flexibility & Reliability Assessments

The Spine Framework allows for the results of the Pathways model to be used for further assessment in other models, including the flexibility and reliability models. The Flexibility model is run for a full continuous year (without representative periods) and with more operational detail. To demonstrate the capability of the framework to carry out flexibility assessments, the Hydrogen Network scenario was evaluated using the flexibility model. This incorporates a long-term model which optimises the salt cavern storage usage which is passed to a rolling operations model allowing the hydrogen to be appropriately valued while allowing the freedom for deviations according to short-term system needs. The flexibility assessment confirmed the capability of the portfolio to meet the system inertia and reserve requirements, with no loss of load.

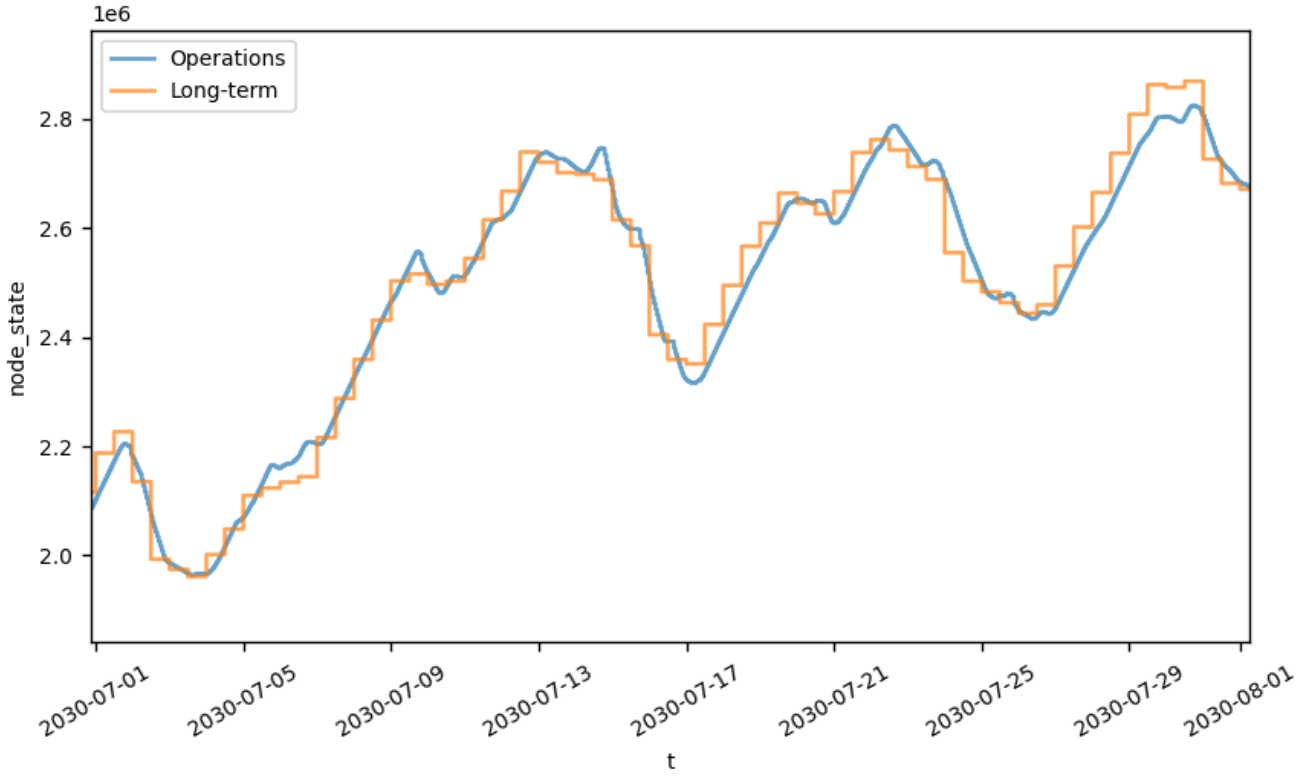


Figure 12 Salt Cavern trajectory for 1 month of operation in both the Long-term stage and the hourly operations model

Figure 12 shows the salt cavern hydrogen storage for 1 month of operation. The trajectories at both the long term and operations stages are closely matched, but the flexibility needs of the system are higher for the hourly

operations model and deviations are required from the coarser long term trajectory.

A state-of-the-art, generalised reliability model was developed specifically for this project which implements resource adequacy assessment suitable for a future integrated energy system with long term storage and high shares of variable renewables. Reliability is assessed for portfolios from the Hydrogen Network scenario, considering 30 weather years and multiple outage patterns while fully considering the contribution of long-term storage. The implementation involved adding monte-carlo capabilities to the SpineOpt energy system modelling framework allowing a wide variety of assessments to be carried out within the framework. The model was employed to evaluate multi-sector resource adequacy analysis of the Hydrogen Network scenario with results highlighting the need to consider resource adequacy at the portfolio optimisation stage. This can be the subject of future work.

5. Discussion & Recommendations

The main focus of the Spine H2-IRL project is developing open source modelling tools, facilitating future modelling of the Irish energy system. A suite of models have been created to address many of the modelling challenges for the future energy transition, including an aggregated 220+ kV electricity network model, a pathways model incorporating 7 scenarios (primarily net zero electricity system), a Flexibility model which allows for more detailed analysis and a reliability Assessment Model.

The scenarios, which draw on established European and Irish scenarios consistent with relevant European and national policy directives and assumptions, explore four core hydrogen scenarios, with different levels of network expansion and different input assumptions for the hydrogen technologies. Three additional scenarios consider *Business as Usual* (without a net-zero target), an *Alternative Net-Zero* scenario, relying on carbon capture and storage, long duration energy storage and negative emission technologies and an *All Options* scenario, with investments in both carbon capture and storage technologies and large scale hydrogen infrastructure.

The detailed, high temporal and geographic resolution results generated by the *Pathways* Model demonstrate the capability of the tools and provide valuable insights for the development of a future net zero electricity system. The Reliability Assessment tool has been developed for very fast run times facilitating large number of runs. This is valuable resource which has been made publicly available, facilitates thorough reliability analysis for the future evolving power system.

Comprehensive sensitivity analysis is beyond the scope of Spine H2-IRL. The scenarios were selected to capture barriers and opportunities for a future hydrogen based electricity system. The Spine H2-IRL tools should be exploited further to complete broader analysis of the Irish energy system.

Key Insights

- Hydrogen technologies have the potential to provide the backbone of a net-zero power system, as well as playing a role in the decarbonisation of the wider energy system.
- To adequately capture the complex interactions across space and time, a modelling approach is required that can simultaneously optimise investments while capturing seasonality of long-term storage and the impact of detailed operations.
- Low-cost hydrogen storage and large-scale infrastructure play an important role in the roll-out of a cost-effective hydrogen-based energy system.
- A hydrogen-based energy system can integrate very large shares of renewable generation and provide stability services for the power system.
- The future system can integrate large shares or renewable generation with limited transmission upgrades when investment decisions are optimised and location specific. This highlights the importance of considering regional resources and networks at the portfolio optimisation stage.
- Very large capacities of renewable generation are required to achieve a net-zero power system based primarily on hydrogen technologies.
- Results from the All Options scenario demonstrate that significant efficiencies can be achieved by providing some energy capacity from alternative sources.
- While the future role of carbon capture and storage is uncertain, other options should be considered for providing baseload and system stability services, including low inertia systems / alternative sources of inertia.
- The Spine H2-IRL open models can be leveraged to consider a broader range of scenarios and sensitivities for future net zero options.

Recommended future work

Generally, the focus of this project was on development of open models along with a complimentary analysis demonstrating how they can be used. The analysis is not intended to be exhaustive, or definitive. It is our hope that the models are further exploited to study the transition further and carry out further analysis. Some specific opportunities for further work are outlined below.

- While a comprehensive multi-sector reliability assessment framework was developed, time did not permit the results of these assessments to be fully accounted for in the pathway modelling phase. Framing of resource adequacy constraints to be included in this phase, informed by reliability assessment can be the subject of future work.
- The modelling outcomes represent a high degree of optimisation of instantaneous production across the network to mitigate constraints and meet demand across multiple sectors. Exploring the operability of the power system with co-optimised dispatch of distributed generation, storage and hydrogen production is an area for further investigation.
- The results show the high value of locationally optimal investment decisions. Further work should explore how this could be realised in practice.
- Dynamics and stability have not been considered, although an inertial floor is included as a constraint. The potential of grid-forming technology and future low-inertia power systems should be studied along with the implications for a hydrogen-based power system.
- The role of interconnection was not extensively explored in this work. Future work could consider alternative interconnection capacities and alternative evolutions of the broader European power system.
- Exploring the development of an established international hydrogen market and the potential role of large scale imports /exports of hydrogen and hydrogen based fuels.
- Additionally, future work could consider broader sensitivity analysis including the following areas
    - Low inertia systems / alternative sources of inertia
    - Available capacities of flexible demand / EV operation
    - Alternative CAPEX assumptions
    - Alternative fuel and carbon price assumptions
    - The integration of heat networks

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